

Research Proposal

Inflation Targeting and Financial Stability in Selected Emerging and Developed Economies, 1995–2022: A Data Science-Based Study Using Four Stability Indicators

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1. Background of the Study

Inflation targeting is a monetary-policy framework in which a central bank publicly announces an inflation target and uses policy instruments, mainly interest rates, to keep inflation close to that target over the medium term. Bernanke and Mishkin (1997) describe inflation targeting as a flexible policy framework rather than a strict mechanical rule. Its key features are transparency, communication, accountability, and central-bank credibility.

Inflation targeting became popular because stable inflation can reduce uncertainty, anchor expectations, support investment decisions, and improve public confidence in monetary policy. Mishkin and Schmidt-Hebbel (2007) show that inflation targeting can improve inflation outcomes and reduce responses to external shocks. However, price stability and financial stability are not the same. The global financial crisis showed that an economy can have relatively low inflation while still facing banking-sector weakness, excessive credit growth, exchange-rate pressure, and asset-price imbalances.

The literature is mixed. Fazio, Tabak, and Cajueiro (2015) find that banking systems in inflation-targeting countries are generally more stable. Mishra and Dubey (2022) also find positive spillover effects in emerging-market economies. In contrast, Fouejieu (2017) finds that inflation-targeting emerging economies may still be financially fragile, while Frappa and Mésonnier (2010) show that inflation targeting can coexist with stronger house-price growth in industrial economies. These findings suggest that inflation targeting should not be treated as automatically stabilizing or destabilizing.

This study keeps the research focus on inflation targeting and financial stability. Because the proposal is submitted for AI Programming, the methodology is data science-based: Python-based machine learning will be used to classify country-year financial-stability risk and to identify patterns across selected emerging and developed economies. The analysis uses four stability indicators: banking stability, credit growth, exchange-rate volatility, and inflation volatility.

2. Problem Statement

An ideal monetary-policy framework should achieve low and stable inflation while also supporting financial stability. In practice, the relationship remains unclear. Some studies find that inflation targeting improves banking stability, while others show that inflation-targeting economies may still experience credit booms, exchange-rate instability, and financial-sector vulnerability. This creates uncertainty for policymakers and analysts because inflation targeting is often evaluated mainly as a price-stability framework, while its wider financial-stability effects remain contested.

A further problem is measurement. Financial stability is multidimensional and cannot be captured well by a single variable. Higher banking stability usually signals lower risk, while higher exchange-rate volatility and inflation volatility signal higher risk. Credit growth is more ambiguous because moderate growth may reflect financial deepening, while unusually rapid growth may signal vulnerability. This study addresses the problem by constructing a financial-stability risk label from four indicators and by using AI/data science methods to classify and interpret risk patterns.

3. Research Gap

The first gap is that previous studies provide mixed and fragmented evidence on inflation targeting and financial stability. Some studies suggest that inflation targeting supports financial stability, while others find that inflation-targeting economies may still experience financial fragility, credit expansion, asset-price pressure, or exchange-rate vulnerability. Therefore, a balanced study is needed that does not assume inflation targeting is uniformly stabilizing or destabilizing.

The second gap concerns measurement. Many studies focus on only one financial-stability channel, such as banking stability, systemic risk, housing prices, or capital flows. Since financial stability is multidimensional, a study that combines several indicators while still interpreting them separately can provide a more complete picture.

The third gap concerns country comparison. Emerging and developed economies differ in institutional quality, financial-market development, exchange-rate exposure, and macroprudential capacity. These differences may shape how inflation targeting relates to financial stability, but they are not always compared clearly within the same research design.

The fourth gap is methodological. Traditional panel regressions are useful for testing average relationships, but they may not fully capture nonlinear patterns, interaction effects, or classification-based financial-stability profiles. A data science approach can complement this literature by predicting high-risk country-year observations and ranking the variables most associated with risk.

The fifth gap is a data-structure issue that must be treated transparently. Several early developed-country inflation targeters adopted inflation targeting before 1995, so inflation-targeting status is constant for those countries during the study period. In a fixed-effects model this would be absorbed by country effects; in machine learning it becomes a near-zero-variance feature for those countries. The proposed sample design partly addresses this issue by including additional developed economies that adopted inflation targeting within the 1995–2022 period, such as Norway, Iceland, South Korea, and Israel. This improves within-sample variation in the developed-economy group, although interpretation of the inflation-targeting variable will still remain cautious (Hammond, 2012).

4. Aim and Objectives

4.1 Aim

The aim of this study is to examine the relationship between inflation-targeting monetary policy and financial-stability risk in selected emerging and developed economies from 1995 to 2022, using a data science-based methodology.

4.2 Objectives

- To explain the relationship between inflation targeting, price stability, and financial stability.
- To construct a country-year dataset using macroeconomic, financial, institutional, and policy indicators.
- To develop a financial-stability risk label using banking stability, credit growth, exchange-rate volatility, and inflation volatility.

- To train and compare machine learning classification models that predict high financial-stability risk.
- To apply K-means clustering and PCA visualization to identify financial-stability profiles across the sample.
- To interpret model results using feature importance and, where feasible, SHAP or LIME.
- To address responsible AI issues, including transparency, small-sample limitations, bias, non-causal interpretation, and GitHub-based reproducibility.

5. Research Questions and Hypotheses

5.1 Research Questions

- What is the relationship between inflation targeting and financial-stability risk in selected emerging and developed economies from 1995 to 2022?
- How does this relationship differ between selected emerging economies and selected developed economies?
- How are banking stability, credit growth, exchange-rate volatility, and inflation volatility associated with inflation-targeting regimes?
- To what extent do institutional quality, macroprudential policy, and country type influence financial-stability risk under inflation-targeting regimes?
- How can AI/data science methods, including classification, clustering, and feature-importance analysis, be used to identify and interpret financial-stability risk patterns?

5.2 Hypotheses

H1: Inflation-targeting regimes are associated with lower financial-stability risk in some dimensions, especially lower inflation volatility and stronger banking stability.

H2: Inflation targeting is expected to show a clearer association with inflation volatility and banking stability than with credit growth and exchange-rate volatility, because credit and exchange-rate outcomes are more strongly affected by external shocks, financial cycles, and country-specific conditions.

H3: Emerging economies are expected to show higher financial-stability risk than developed economies in some country-year observations because of external shocks, exchange-rate exposure, and institutional constraints.

H4: Higher institutional quality and stronger macroprudential policy are expected to be associated with lower financial-stability risk.

H5: AI/data science methods are expected to identify meaningful financial-stability risk patterns and outperform a simple majority-class baseline, but the results will be interpreted as exploratory, predictive, and associative rather than causal.

These hypotheses keep the research focus on inflation targeting and financial stability. AI/data science is used as the analytical method, not as a separate research topic.

6. Brief Literature Review

Foundational studies explain why inflation targeting can improve credibility, transparency, and inflation expectations. Bernanke and Mishkin (1997) provide the theoretical foundation, while Mishkin and Schmidt-Hebbel (2007) show that inflation targeting can improve inflation outcomes but does not always outperform strong non-targeting regimes. Fraga, Goldfajn, and Minella (2003) add that emerging economies face stronger shocks, lower credibility, and weaker institutions, making their inflation-targeting experience different from that of developed economies.

Evidence on financial stability is mixed. Fazio et al. (2015) find that inflation-targeting banking systems are generally more stable, supporting the inclusion of banking stability as one indicator. Fouejieu (2017), however, finds that inflation-targeting emerging economies may still be financially fragile, while Mishra and Dubey (2022) find positive spillover effects to banking resilience and capital inflows. These contrasting results support a multi-indicator and country-comparative approach.

Other studies show why financial stability should not be reduced to inflation stability. Frappa and Mésonnier (2010) find that inflation targeting can be associated with stronger house-price growth, while Akram and Eitrheim (2008) show that responses to asset prices and credit can have mixed effects. These studies support examining credit growth, exchange-rate volatility, and inflation volatility separately.

Institutional quality and macroprudential policy are also important. Fazio et al. (2018) show that institutional quality conditions the effect of inflation targeting on financial stability. Kim and Mehrotra (2017) show that monetary and macroprudential objectives can create trade-offs, while Mugrabi et al. (2026) find that macroprudential tools may work more effectively under inflation targeting. Ngepah et al. (2026) further support comparing emerging and developed economies because developed economies show stronger financial-stability survival probabilities, while emerging economies remain more exposed to external volatility. Woodford (2012) argues that inflation targeting can be made flexible enough to incorporate financial-stability concerns rather than being abandoned.

6.1 Summary of Literature

Table 1: Summary of Reviewed Literature

Study group	Key finding	Relevance to this proposal
Bernanke & Mishkin (1997); Mishkin & Schmidt-Hebbel (2007)	Inflation targeting improves credibility and inflation performance, but success is conditional.	Provides theoretical foundation and supports cautious interpretation.
Fraga et al. (2003)	Emerging economies face stronger shocks, weaker institutions, and lower credibility.	Supports emerging-versus-developed comparison.
Fazio et al. (2015); Mishra & Dubey (2022)	Inflation targeting may improve banking stability and capital-flow resilience.	Supports banking-stability and supportive-evidence channels.

Fouejieu (2017)	Inflation-targeting emerging economies may still be financially fragile.	Provides cautionary evidence and motivates risk classification.
Frappa & Mesonnier (2010); Akram & Eitrheim (2008)	Low inflation may coexist with housing, credit, and asset-price risks.	Supports multi-indicator measurement.
Fazio et al. (2018); Kim & Mehrotra (2017); Mugrabi et al. (2026)	Institutional quality and macroprudential policy condition financial-stability outcomes.	Supports institutional and policy features in the model.
Ngepah et al. (2026); Woodford (2012)	Country type matters, and inflation targeting can be adapted to include financial-stability concerns.	Supports comparative and flexible-policy framing.

Overall, the literature shows that inflation targeting is neither uniformly stabilizing nor uniformly destabilizing. Its association with financial stability depends on the country group, institutional quality, macroprudential policy, timing of adoption, and the specific indicator examined. This proposal builds on that evidence by applying AI/data science methods to classify and interpret multidimensional financial-stability risk.

7. Conceptual Framework

The conceptual framework keeps inflation targeting and financial stability as the central research relationship. Inflation targeting is the main monetary-policy concept, while financial stability is represented through four indicators: banking stability, credit growth, exchange-rate volatility, and inflation volatility. Institutional quality, macroprudential policy, and country type are treated as contextual factors that may moderate the relationship.

Inflation targeting may support financial stability through stabilising mechanisms such as central-bank credibility, anchored expectations, and lower inflation uncertainty. However, it may also fail to prevent risk if credit expansion, leverage, exchange-rate pressure, or financial risk-taking increase while consumer-price inflation remains stable. Therefore, the study expects the relationship to vary across indicators and country groups.

Figure 1. Conceptual Framework Linking Inflation Targeting and Financial Stability, Analyzed Using an AI/Data-Science Methodology

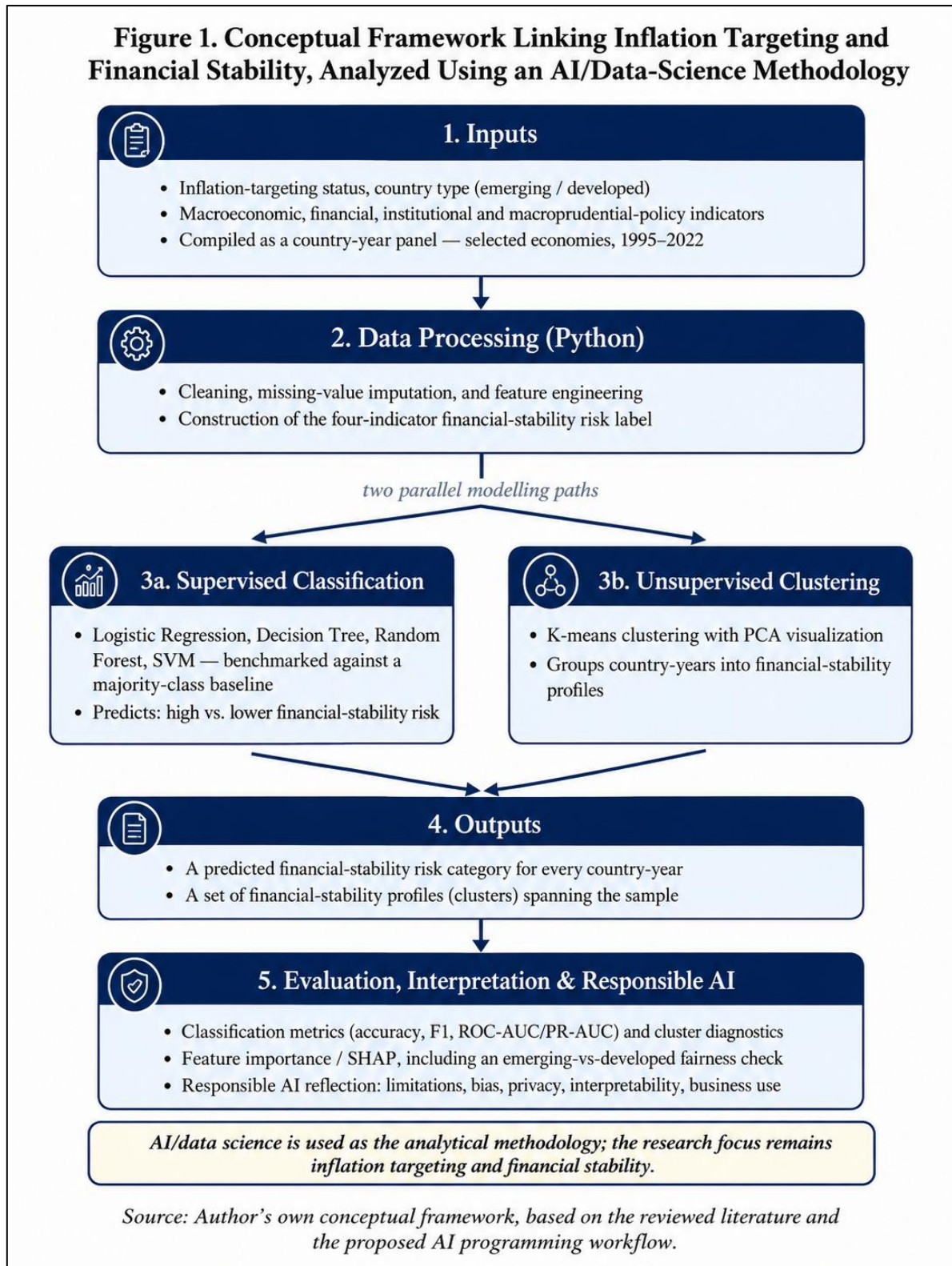


Figure 1 shows that secondary country-year data will be processed in Python, used to construct a four-indicator financial-stability risk label, and analyzed through two paths: supervised classification and unsupervised clustering. The outputs will be evaluated, explained, and interpreted responsibly as predictive and associative evidence, not as causal proof.

8. Methodology

8.1 Research Design, Sample, and Data Sources

The study uses a quantitative AI/data science research design. The unit of analysis is a country-year observation. The proposed sample will cover 1995–2022 and include up to 20 selected economies, with approximately ten emerging economies and ten developed economies where data availability permits. This sample supports the data science methodology by increasing the number of country-year observations available for classification, clustering, validation, and feature-importance analysis. With 20 countries over 28 years, the maximum sample would be about 560 country-year observations before missing data are considered.

The proposed emerging economies may include Brazil, Mexico, South Africa, Indonesia, Turkey, Chile, Colombia, Peru, Poland, and the Czech Republic. The proposed developed economies may include Canada, the United Kingdom, Sweden, Australia, New Zealand, Norway, Iceland, South Korea, Israel, and another suitable advanced inflation-targeting economy depending on data availability. The developed-economy selection also improves within-sample variation in inflation-targeting adoption because Norway and Iceland adopted inflation targeting in 2001, South Korea in 1998, and Israel in 1997, all within the 1995–2022 study period (Hammond, 2012). The final sample will be confirmed after checking whether the required macroeconomic, financial, institutional, and policy data are sufficiently available and comparable. If some countries have excessive missing data for key indicators, the final sample may be smaller than 20 countries, and the exclusion criteria will be documented clearly.

Secondary data will be collected from reliable international sources such as IMF databases, World Bank World Development Indicators, World Bank Worldwide Governance Indicators, Bank for International Settlements data, central-bank reports, and macroprudential policy databases. The data will be compiled into a structured country-year dataset before cleaning and analysis. The dataset will include inflation-targeting status, country type, macroeconomic variables, institutional-quality indicators, macroprudential-policy indicators, and financial-stability indicators.

A remaining data-availability issue will be reported transparently. Macroprudential-policy data may be less complete for some smaller or later-included economies, meaning that the final sample may become unbalanced across variables and countries. Therefore, the final dataset will document missingness, variable coverage, and any country exclusions before model training.

8.2 Variables and Target Construction

Variable group	Examples	Purpose
Country information	Country, year, emerging/developed status	Identify observations and group differences.
Policy and institutions	Inflation-targeting dummy, institutional quality, macroprudential policy	Capture monetary-policy status and contextual moderators.
Macroeconomic features	GDP growth, interest rate, trade openness, public debt, financial openness	Provide predictive context.
Stability indicators	Banking stability, credit growth, exchange-rate volatility, inflation volatility	Construct the risk label and clustering profiles.

The main supervised-learning target is a binary label: High Financial-Stability Risk = 1 and Lower Financial-Stability Risk = 0. The label will be constructed by standardizing the four indicators and aligning their signs so that higher values represent greater risk. For example, banking stability will be reversed if a higher value originally means stronger stability.

A composite risk score will then be created, and the top 25% or 30% of observations will be labelled high risk. A quantile-based threshold is used because complete and comparable crisis-event labels may not be available for all selected countries and years. A domain-based crisis threshold would be preferable if a consistent crisis dataset were available, but the selected indicators capture different forms of financial-stability pressure rather than one official crisis definition. Therefore, the quantile approach provides a practical and transparent way to identify relatively high-risk observations within the sample. The threshold will be documented clearly and tested with at least one alternative cut-off.

8.3 Feature Engineering and Leakage Control

Categorical variables will be encoded, numerical variables will be scaled where required, and missing values will be handled through documented imputation. To avoid preprocessing leakage, imputation and scaling will be fitted only on the training data and applied to the test data through a scikit-learn pipeline.

The same four indicators used to construct the risk label will not be used as same-year predictors in the classification models. Otherwise, the model would simply reconstruct its own target. If a stability indicator is used as a predictor, a lagged prior-year value will be used instead. This restriction does not apply to clustering, where the indicators are used directly to group observations into financial-stability profiles.

8.4 Machine Learning Methods, Evaluation, and Explainability

The supervised-learning stage will train Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine models, benchmarked against a majority-class baseline. The unsupervised-learning stage will use K-means clustering, with PCA visualization to support interpretation. The final interpretation of clusters will be based on original features rather than PCA plots alone.

Model evaluation will use accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC, and PR-AUC where appropriate. Because the dataset is still relatively small for machine learning, the results will be treated as exploratory and illustrative rather than as statistically powerful predictions. Model performance will not be judged using only one train/test split. Instead, the study will compare results across validation approaches where feasible, including country-grouped validation and a time-based split. Evaluation results will be reported cautiously, using metric ranges or repeated-validation results where possible, rather than relying only on a single point estimate. Feature-importance results will also be interpreted carefully because rankings may be unstable in a small sample.

Model interpretation will use tree-based feature importance and, where feasible, SHAP or LIME. The study will also check whether performance differs between emerging and developed economies, such as through recall or false-positive-rate differences. Results will be presented as predictive and associative, not causal.

8.5 Reproducibility

The workflow will be implemented in Python using Jupyter Notebook or Google Colab and documented in a GitHub repository. The repository will include the notebook, dataset source links or citations, README file, outputs, evaluation metrics, visualizations, and reproducibility instructions. Restricted or licensed data will not be uploaded directly to GitHub.

9. Expected Contribution

Academically, this study connects the inflation-targeting and financial-stability literature with machine-learning-based risk classification. It treats financial stability as multidimensional and examines how inflation targeting, institutions, macroprudential policy, and country type relate to risk patterns.

Practically, the study may provide a screening tool for analysts, policymakers, treasury teams, and financial institutions assessing cross-country macro-financial exposure. The model is not intended to replace expert judgement, but it can organize information and highlight high-risk country-year observations for closer review.

Methodologically, the study contributes a reproducible Python workflow that combines supervised learning, unsupervised learning, model evaluation, explainability, and responsible AI reflection within a finance and policy research question. This includes documenting the data sources, feature construction, risk-label construction, validation approach, model outputs, and limitations in a GitHub-based workflow. This contribution is practical and methodological rather than a claim that machine learning replaces economic theory or causal analysis.

10. Ethical Considerations and Responsible AI

The study uses secondary country-level macroeconomic and financial data. It does not involve human participants, surveys, interviews, experiments, or personal information, so consent and personal anonymity are not directly applicable. However, responsible data use remains important.

All data and academic sources will be cited, variable definitions and missing-data treatment will be documented, and the model will not be presented as a causal or final decision-making tool. The high-risk label depends on a threshold choice, so at least one alternative threshold will be tested. If licensed or restricted data are used, the repository will provide source information and reproduction instructions without uploading restricted material.

The model could create misleading interpretations if emerging economies are automatically viewed as unstable. Therefore, classification performance will be checked across emerging and developed groups, and results will be interpreted with context. Human expert review is necessary before model outputs are used in business or policy decisions.

11. Limitations of the Study

- Financial stability is difficult to measure with one label, and the high-risk threshold may affect which observations are classified as risky.

- Machine learning identifies predictive patterns, not causal effects. The study cannot prove that inflation targeting directly causes financial stability or instability.
- Although the sample may include up to 20 economies, it may still be relatively small for machine learning after missing data are considered. Therefore, model performance and feature importance will be interpreted cautiously.
- Some developed economies adopted inflation targeting before 1995, so the inflation-targeting variable has limited within-sample variation for those countries. The proposed sample design partly addresses this by including developed economies with within-sample adoption variation.
- The country sample is partly shaped by data availability. Although the selected countries are relevant inflation-targeting cases and allow an emerging-versus-developed comparison, the findings should be interpreted as evidence from a focused comparative sample rather than as fully generalizable conclusions for all countries.
- Data availability may vary across countries and years, especially for macroprudential and banking-sector indicators. The sample may become unbalanced across countries and variables if some indicators are missing more often for smaller or later-included economies.
- Credit growth must be interpreted carefully because it can represent either financial deepening or financial vulnerability.

12. Proposed Timeline

Table 3: Proposed Research Timeline

Stage	Main activity	Expected completion
Stage 1	Topic selection, background, problem statement, research gap, aim, objectives, and research questions.	Completed
Stage 2	Data science-based proposal with literature review, conceptual framework, methodology, contribution, ethics, timeline, and references.	Current stage
Data Collection	Collect and screen data for up to 20 selected economies; finalize the usable sample based on data availability.	After proposal feedback
Data Preparation	Clean data, handle missing values, construct features, build the risk label, and document the threshold.	Following data collection
Model Development	Train classification models and K-means clustering model.	Following data preparation
Model Evaluation	Evaluate classification metrics, cluster diagnostics, PCA visualization, explainability, and group-performance checks.	After model training
Final Paper and GitHub	Finalize paper, notebook, README, outputs, Responsible AI reflection, and GitHub submission.	End of term

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